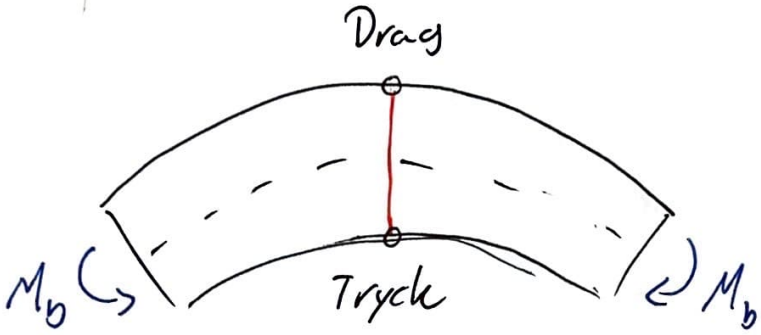
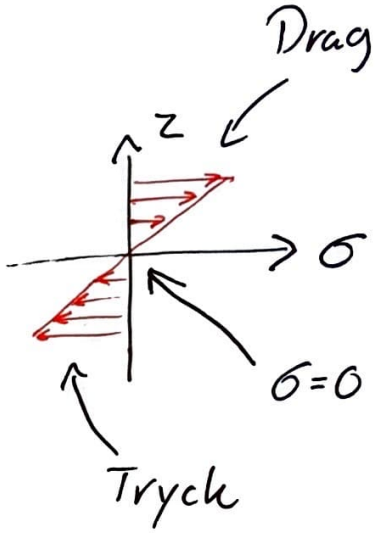
Övning 8 - Böjning av balk

2,4: 37, 39, 42

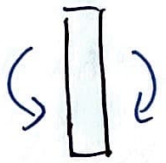




Geometri spelar stor roll



lättböjd



svårböjd

Uttrycks med I_y

Normalspänning vid böjning

$$\sigma = \left(\frac{M}{I} z\right) + \left(\frac{N}{A}\right)$$

Olika beroende på vart i tvärsnittet!

Bidrag från böjning

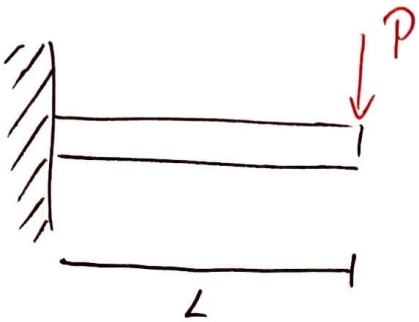
Bidrag från normalkraften

$$\sigma_{Max} = \frac{|M|}{W_b}$$ [FS:6.7-8]

2.4.37

Givet: $P = 1500\,N$

$L = 2\,m$





Sökt: a) D så att $\sigma_s = 20\,MPa$ inte orsakar plasticering

b) Inkludera egenvikt av balken

$\rho_{trä} = 600\,kg/m^3$

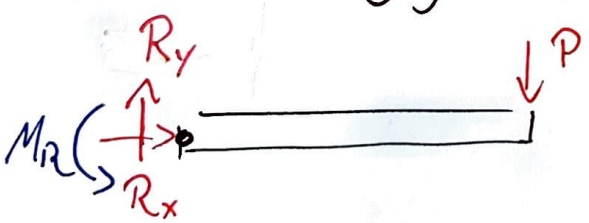
c) Dimensionera en IPE-balk

$\rho_{stål} = 7840\,kg/m^3$

$\sigma_s = 100\,MPa$

---

a) ① Frilägggning

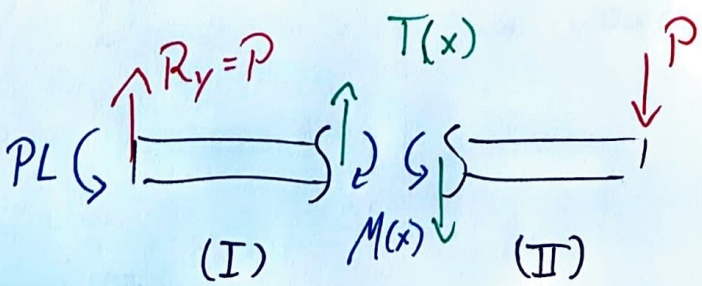


$\uparrow: R_y - P = 0 \quad R_y = P$

$\rightarrow: R_x = 0$

$\circlearrowright: -M_R + PL = 0 \quad M_R = PL$

② Snittning



$$I = \frac{\pi \left(\frac{D}{2}\right)^4}{4} = \frac{\pi D^4}{64}$$

Jämvikt snittning

$\rightarrow: N(x) = 0$

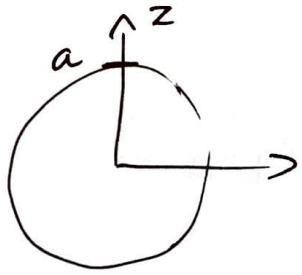
$\uparrow: T(x) + P = 0 \qquad T(x) = -P$

$\circlearrowleft: M(x) + Px - PL = 0 \qquad M(x) = P(L-x)$

③ Spänning

$$\sigma = \frac{M(x)}{I} z + \frac{N}{A} = \frac{P(L-x)}{\frac{\pi D^4}{64}} z = \frac{64P(L-x)}{\pi D^4} z$$

Max fås där z är som störst



$$z = a = \frac{D}{2}$$

$$\sigma_{max} = \frac{32P(L-x)}{\pi D^3} \xrightarrow{x=0} \frac{32PL}{\pi D^3}$$

Dimensioneringsvillkor

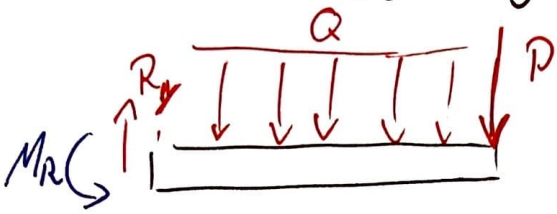
$\sigma_{max} = \sigma_s$

$$\sigma_s = \frac{32PL}{\pi D^3} \qquad D = \sqrt[3]{\frac{32PL}{\pi \sigma_s}} = 0.1152\,m$$

Diametern ska vara 11.5cm

2.4.37 del 2

b) ① Frilägging



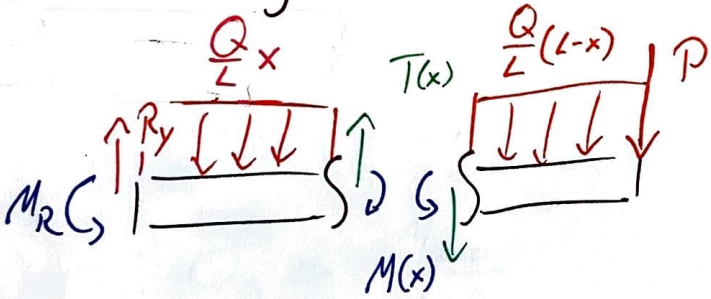
$$Q = \rho g V = \rho g a^2 \pi L$$

Jämvikt

$$\uparrow: R_y - Q - P = 0 \qquad R_y = P + \rho g \pi a^2 L$$

$$\circlearrowright: -M_R + Q\frac{L}{2} + PL = 0 \qquad M_R = \frac{\rho g \pi a^2 L^2}{2} + PL$$

② Snittning



$$\uparrow: R_y - \frac{Q}{L}x + T(x) = 0 \qquad T(x) = \rho g \pi a^2 x - P - \rho g \pi a^2 L$$

$$\circlearrowright: M(x) - M_R + R_y x - \frac{Q}{L}x\frac{x}{2} = 0 \qquad M(x) = \frac{\rho g \pi a^2}{2}(L-x)^2 + P(L-x)$$

③ spänning

$$\sigma(x) = \frac{M(x)}{I_y} z \qquad I_y = \frac{\pi D^4}{64}$$

$$M_{max} = M(0) = \frac{\rho g \pi a^2 L^2}{2} + PL$$

$$z_{max} = a = \frac{D}{2}$$

$$\Rightarrow \sigma_{max} = \frac{64\left(\frac{\rho g \pi (\frac{D}{2})^2 L^2}{2} + PL\right)}{\pi D^4} = 20.8\,MPa$$

2.4:37 del 4

c) Dimensionera en IPE-balk

$$M_{max} = \frac{\rho g \pi a^2 L^2}{2} + PL$$

byt ut till ny area A

$$M_{max} = \frac{\rho g A L^2}{2} + PL$$

$$\sigma_{max} = \frac{M_{max}}{W_b} \quad [FS.\ 6.8]$$

Dimensioneringskrav

$$\sigma_s > \sigma_{max}$$

$$\sigma_s \geq \frac{M_{max}}{W_b} \Rightarrow W_b > \frac{M_{max}}{\sigma_s}$$

$$W_b > \frac{\frac{\rho g A L^2}{2} + PL}{\sigma_s}$$

insatta värden

$$\Rightarrow W_b > 3 \cdot 10^{-5} + 1.54 \cdot 10^{-3}$$

Testa olika geometrier [FS Tabell 30.5]

IPE 80 $W_b = 20 \cdot 10^3\ mm^3$ $A = 762\ mm^2$

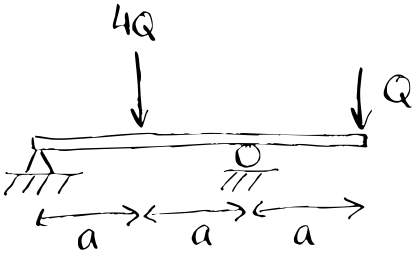
$20 \cdot 10^3 \cdot 10^{-9} > 3 \cdot 10^{-5} + 1.54 \cdot 10^{-3} \cdot 762 \cdot 10^{-6}$

$20 \cdot 10^{-6} > 31.2 \cdot 10^{-6}$ (Fel!)

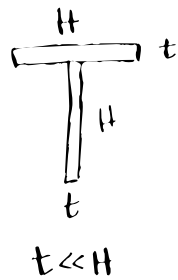
IPE 100 $W_b = 34.2 \cdot 10^3\ mm^3$ $A = 1032\ mm^2$

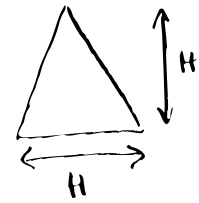
$34.2 \cdot 10^{-6} > 31.6 \cdot 10^{-6}$ (Fungerar!)

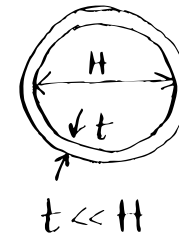
2.4.39



3 tvärsnitt:

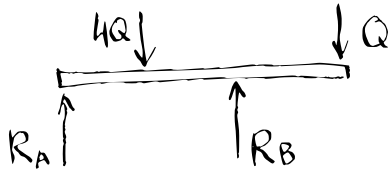






Sökt: σ_{max} och σ_{min} för de olika tvärsnitten

① Frilägg, snitta, jämvikt



$$R_A + R_B = 5Q$$

$$4Qa - 2R_B a + 3Qa = 0$$

$$\Rightarrow \begin{cases} R_A = \frac{3}{2}Q \\ R_B = \frac{7}{2}Q \end{cases}$$

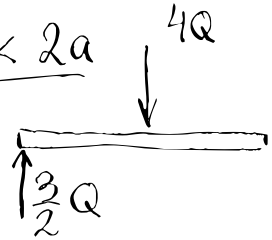
Snitt:

$0 < x < a$



$$M(x) = -\frac{3}{2}Qx$$

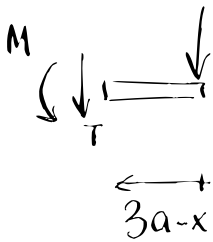
$a < x < 2a$



$$M(x) + \frac{3}{2}Qx - 4Q(x-a) = 0$$

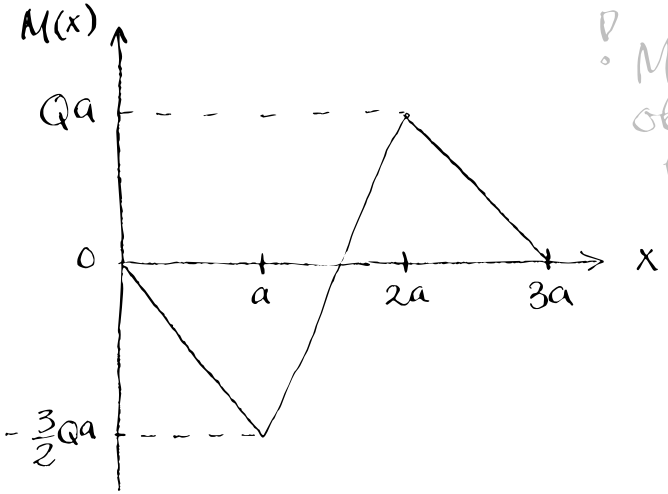
$$\Rightarrow M(x) = \frac{5}{2}Qx - 4Qa$$

$2a < x < 3a$



$M = Q(3a - x)$

Sammanfatta i M-diagram:



! Momentet är oberoende av tvärsnittet

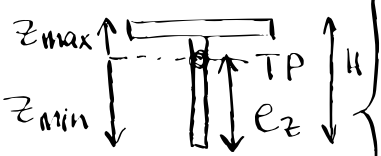
(2) Spänningar

$$\sigma = \frac{M(x)}{I} z \quad [FS\ 6.7]$$

Vi vill maximera och minimera
$\Rightarrow$ Vi kollar på $\max(M(x))$ och $\min(M(x))$.

$$\begin{cases} M_{max} = Qa \\ M_{min} = -\frac{3}{2}Qa \end{cases}$$

③ I – yttröghetsmoment $[m^4]$, z_{min}, z_{max}



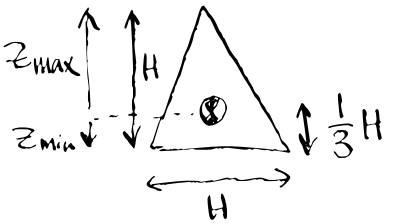
tab 30.2.10 (31.2.10)

$I_y = \ldots$, där $e_z = H\frac{Ht + Ht\frac{1}{2}}{Ht+Ht} = \frac{3H}{4}$

$\Rightarrow \ldots \Rightarrow I_y = \frac{5tH^3}{24}$

$z_{min} = -e_z = -\frac{3H}{4}$

$z_{max} = H - e_z = \frac{H}{4}$

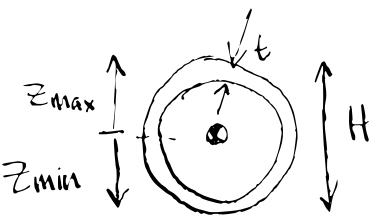


tab 30.1.2 (31.1.2)

$I_y = \frac{bh^3}{36} = \frac{H^4}{36}$

$z_{min} = -\frac{1}{3}H$

$z_{max} = \frac{2}{3}H$



tab 30.1.4 (31.1.4)

$I_y = \pi a^3 t = \pi\frac{H^3}{8}t$

$z_{min} = -H/2$

$z_{max} = H/2$

(4) σ_{max} och σ_{min}

$$\sigma = \frac{M}{I} z$$

Vi har två möjliga M och 2 möjliga z – testa alla 4 kombinationer!

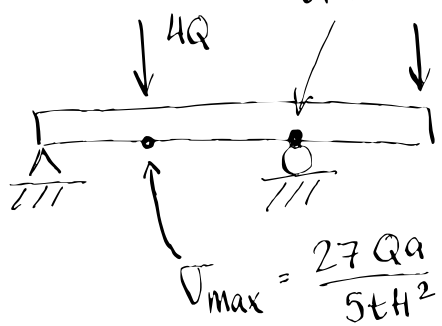
För T-tvärsnittet:



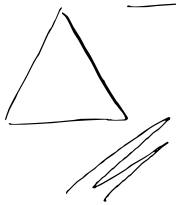
$$I = \frac{5tH^3}{24}$$

	vid $x = a$, $M = -\frac{3}{2}Qa$	vid $x = 2a$, $M = Qa$
På ovansidan $z = H/4$	$\sigma = -\frac{9}{5}\frac{Qa}{tH^2}$	$\sigma = \frac{6Qa}{5tH^2}$
På undersidan $z = -3H/4$	$\sigma = \frac{27Qa}{5tH^2}$	$\sigma = -\frac{18}{5}\frac{Qa}{tH^2}$

Alltså: $\sigma_{min} = -\frac{18}{5}\frac{Qa}{tH^2}$

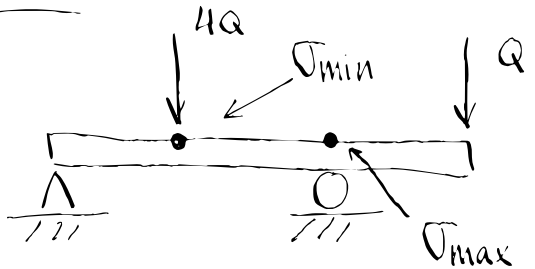


För triangeln:

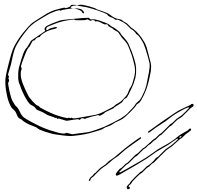


	vid $x = a$ $M = -\frac{3}{2}Qa$	vid $x = 2a$ $M = Qa$
På ovansidan $z = H/4$	$\sigma = -\frac{36Qa}{H^3}$	$\sigma = \frac{24Qa}{H^3}$
På undersidan $z = -3H/4$	$\sigma = \frac{18Qa}{H^3}$	$\sigma = -\frac{12Qa}{H^3}$

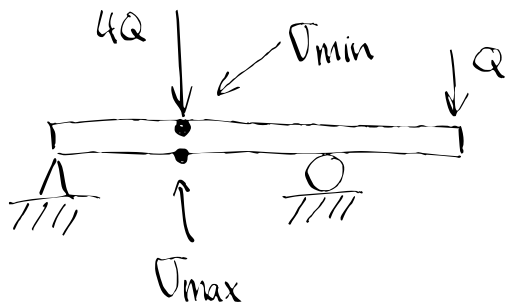
Alltså



För rör:



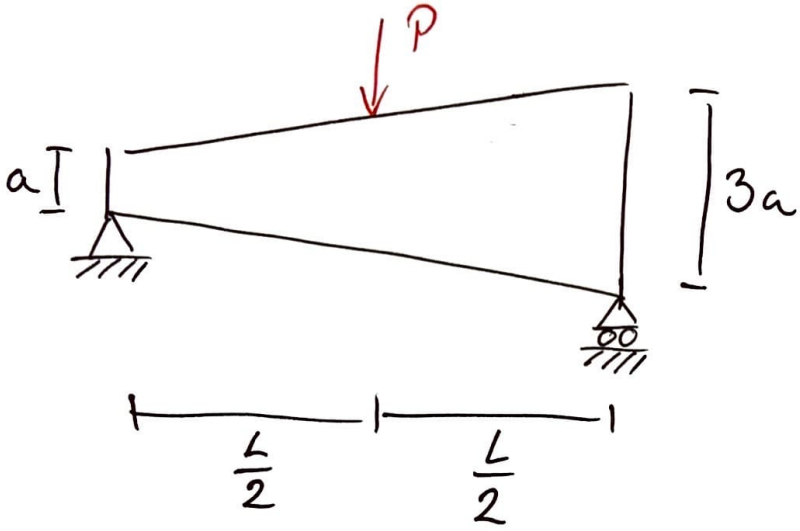
	vid $x = a$ $M = -\frac{3}{2}Qa$	vid $x = 2a$ $M = Qa$
På ovansidan $z = H/4$	$\sigma = -\frac{3Qa}{2\pi H^2 t}$	$\sigma = \frac{Qa}{\pi H^2 t}$
På undersidan $z = -3H/4$	$\sigma = \frac{3Qa}{2\pi H^2 t}$	$\sigma = -\frac{Qa}{\pi H^2 t}$



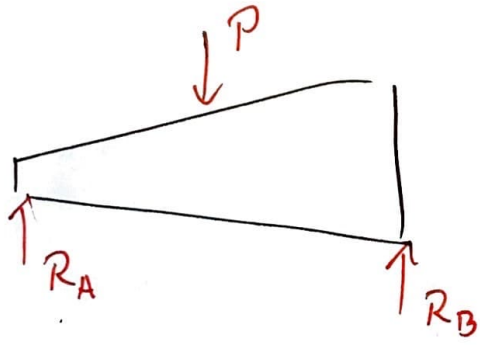
2.4:42

Givet: a, P, L

Sökt: σ_{max} och x-koordinaten där denna finns.



① Frilägging

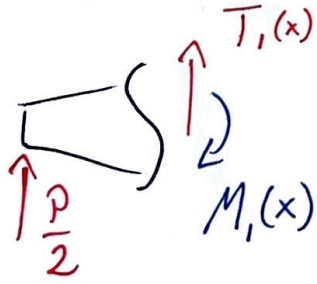


$\uparrow: R_A + R_B - P = 0$

$A: P\frac{L}{2} - R_B L = 0$

$$\Rightarrow \boxed{\begin{aligned} R_B &= \frac{P}{2} \\ R_A &= \frac{P}{2} \end{aligned}}$$

② Snittning

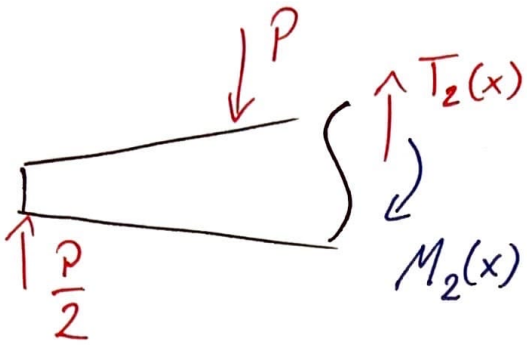


$\uparrow: T_1(x) + \frac{P}{2} = 0$

$$T_1(x) = -\frac{P}{2}$$

$\circlearrowright: M_1(x) + \frac{P}{2}x = 0$

$$M_1(x) = -\frac{P}{2}x$$

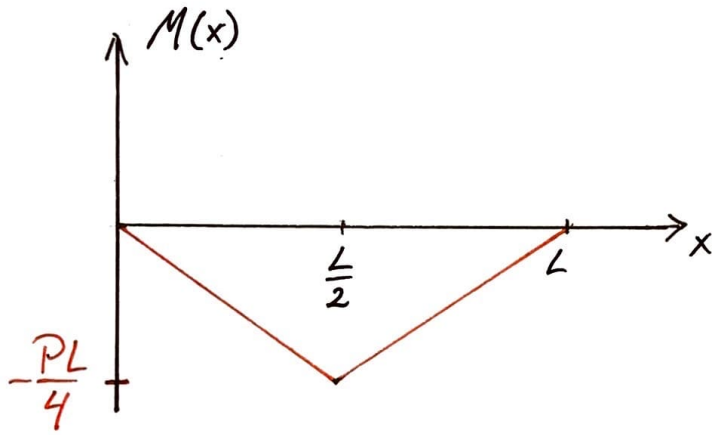


$$\uparrow: \; T_2(x) + \frac{P}{2} - P = 0$$

$$T_2(x) = \frac{P}{2}$$

$$\circlearrowleft: \; M_2(x) + \frac{P}{2}x - P\left(x - \frac{L}{2}\right) = 0$$

$$M_2(x) = \frac{P}{2}x - \frac{P}{2}L$$



$$\sigma_{max,1} = \frac{M_1(x)}{W_b(x)} \qquad \sigma_{max,2} = \frac{M_2(x)}{W_b(x)}$$

$$W_b(x) = \frac{\pi r^3}{4}$$

$$W_b(x) = \frac{\pi}{4}\left(\frac{a}{2}\right)^3\left(1 + \frac{2x}{L}\right)^3$$

$$r(x) = kx + m$$

$$r(0) = \frac{a}{2} \Rightarrow m = \frac{a}{2}$$

$$r(L) = \frac{3a}{2} \Rightarrow k = \frac{a}{L}$$

$$r(x) = \frac{a}{2}\left(1 + \frac{2x}{L}\right)$$

Identifiera x genom derivering av σ_{max}

$$\frac{d|\sigma_{max}|}{dx} = 0$$

$$\frac{d}{dx}\left(\frac{\frac{P}{2}x}{\frac{\pi}{4}\left(\frac{a}{2}\right)^3\left(1 + \frac{2x}{L}\right)^3}\right) = \frac{d}{dx}\left(\frac{x}{\left(1 + \frac{2x}{L}\right)^3}\right) = 0$$

$$\frac{d}{dx}\left(\frac{x}{\left(1+\frac{2x}{L}\right)^3}\right) = \frac{1}{\left(1+\frac{2x}{L}\right)^3} - \frac{3x}{\left(1+\frac{2x}{L}\right)^4}\frac{2}{L}$$

$$\Rightarrow \quad L\left(1+\frac{2x}{L}\right) - 6x = 0$$

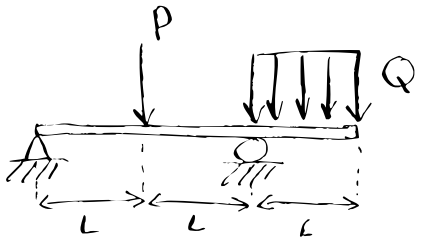
$$L + 2x - 6x = 0$$

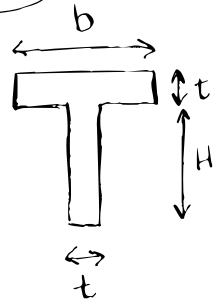
$$\boxed{x = \frac{L}{4}}$$

$$\sigma_{max,1}\left(\frac{L}{4}\right) = \frac{M_1\left(\frac{L}{4}\right)}{W_b\left(\frac{L}{4}\right)} = \frac{\frac{P}{2}\frac{L}{4}}{\frac{\pi}{4}\left(\frac{a}{2}\right)^3\left(1+\frac{2}{L}\frac{L}{4}\right)^3}$$

$$\boxed{\sigma_{max1}\left(\frac{L}{4}\right) = \frac{32PL}{27\pi a^3}}$$

2.4.47

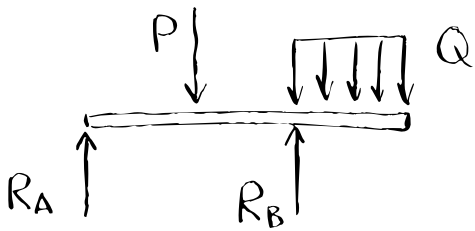




Givet:
$L = 0{,}5$ m
$Q = 30$ kN
$P = 40$ kN
$H = 8$ cm
$b = 12$ cm
$t = 2$ cm

Sökt: σ_{min}, σ_{max}

① Frilägg, snitta, jämvikt

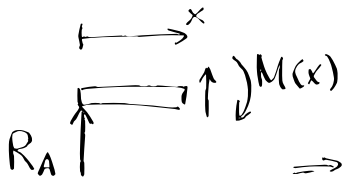


Jämvikt ger:

$$\begin{cases} R_A = \frac{P}{2} - \frac{Q}{4} \\ R_B = \frac{P}{2} + \frac{5Q}{4} \end{cases}$$

Snitt

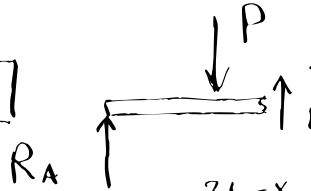
$0 < x < L$



$$M(x) = -R_A x$$

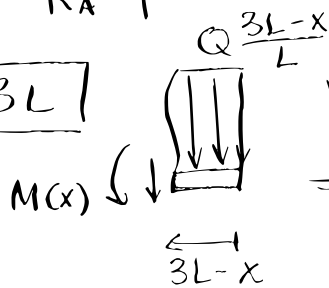
$$\Rightarrow M(x) = \left(\frac{Q}{4} - \frac{P}{2}\right) x$$

$L < x < 2L$



$$\dots M(x) = \frac{Px}{2} + \frac{Qx}{4} - PL$$

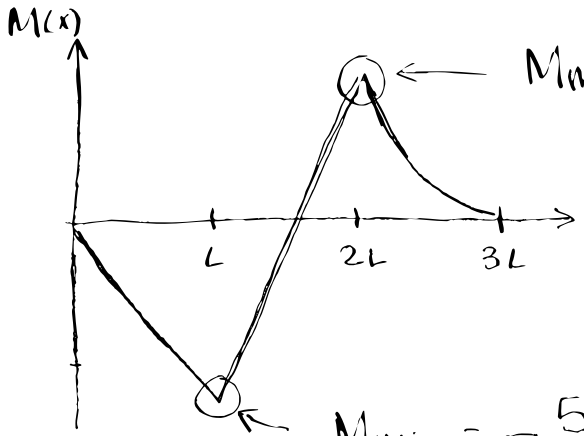
$2L < x < 3L$



$$\circlearrowleft: M(x) - Q\frac{(3L-x)}{L}\frac{(3L-x)}{2} = 0$$

$$\Rightarrow M(x) = \frac{Q}{2L}(3L-x)^2$$

M-diagram med insatta värden:

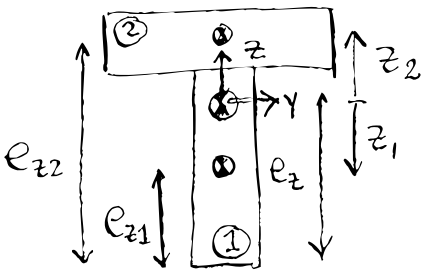


$M_{max} = \frac{3PL}{8}$

$M_{min} = -\frac{5PL}{16}$

Tvärsnitt oberoende av x
$\Rightarrow$ σ_{max}, σ_{min} måste ske vid M_{max} eller M_{min}.

(2) I_y och z



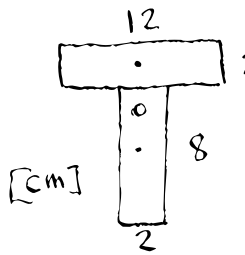
Steiners satsats

$$I_y = \sum_i I_{yi} + z_i^2 A_i \quad [FS\ 30.5]$$

$$\Rightarrow I_y = I_1 + z_1^2 A_1 + I_2 + z_2^2 A_2$$

$$[FS\ 30.2]\quad e_z = \frac{\sum_i e_{zi} A_i}{A} = \frac{e_{z1}A_1 + e_{z2}A_2}{A_1 + A_2}$$

$e_{z1} = 4\,cm$, $e_{z2} = 9\,cm$, $A_1 = 16\,cm^2$, $A_2 = 24\,cm^2$



$$\Rightarrow e_z = \frac{4 \cdot 16 + 9 \cdot 24}{16 + 24} = 7\,cm$$

$$\Rightarrow \begin{cases} z_1 = e_{z1} - e_z = -3\,cm \\ z_2 = e_{z2} - e_z = 2\,cm \end{cases}$$

[FS tab 30.1.1]

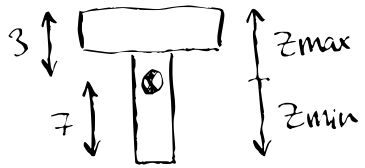
$$I_{rektangel} = \frac{bh^3}{12} \Rightarrow \begin{cases} I_1 = \dfrac{2 \cdot 8^3}{12} \text{ cm}^4 \\ I_2 = \dfrac{12 \cdot 2^3}{12} \text{ cm}^4 \end{cases}$$

Allt känt!

$$\Rightarrow I_y = \frac{2 \cdot 8^3}{12} + (-3)^2 \cdot 16 + \frac{12 \cdot 2^3}{12} + 2^2 \cdot 24 \text{ [cm}^4\text{]}$$

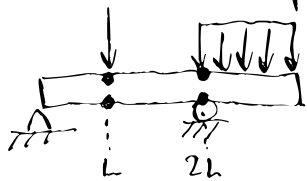
$$\Rightarrow I_y = 333{,}3 \text{ cm}^4$$

[cm]



$$\begin{cases} z_{max} = 3 \text{ cm} \\ z_{min} = -7 \text{ cm} \end{cases}$$

③ Maximala spänningar



← vi är intresserade av dessa fyra punkter.

$\sigma = \frac{M}{I_y} z$	Ovansida $z = 3$ cm	Undersida $z = -7$ cm	
$x = L$, $M = -\frac{5PL}{16}$	$\sigma = -56$ MPa	$\sigma = 131$ MPa	← max drag
$x = 2L$, $M = \frac{3PL}{8}$	$\sigma = 67{,}5$ MPa	$\sigma = -157{,}5$ MPa	← max tryck